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4. Create a visual representation in Figma highlighting the key elements of mobile network protocol stacks.

Aim

To create a visual representation in Figma highlighting the key elements of mobile network protocol stacks and understanding the functions of different protocol layers.

Procedure

1. Open Figma and create a new design file.
2. Create a single page with the title “ Key Elements of Mobile Network Protocol Stacks.”
3. Draw five horizontal rectangular blocks representing Application, Transport, Network, Data Link, and Physical layers.
4. Add protocols and functions to each layer: HTTP/HTTPS, DNS, DHCP; TCP, UDP; IP, ICMP; Wi-Fi MAC, Ethernet; and Wi-Fi radio and cellular radio interface.
5. Use different colors to distinguish the layers and add arrows to show transmission and reception.
6. Create a simple mobile network diagram showing Mobile Device → Wireless Radio Link

→ Base Station → Core Network → Internet/Server.

7. Add a reference section describing LTE, 5G NR, GSM, and the functions of PHY, MAC, RLC, PDCP, RRC, and NAS.
8. Add a data-flow summary and legend. Align all elements and check the diagram for readability and technical accuracy.

01 - Protocol Stack Introduction

MOBILE COMPUTING 01 / 04

THE BIG PICTURE

Mobile Network Protocol Stack

Understanding how data travels through mobile networks.

FROM DEVICE TO INTERNET Data path



Application Layer
Supports services such as web browsing and messaging.

Transport and Network Layers
Help deliver data between applications and devices.

Radio and Link Layers
Carry data across the wireless connection.

WHAT IS A PROTOCOL STACK?

A protocol stack is a set of communication protocols arranged in layers. Each layer performs specific tasks to enable devices and networks to communicate.

Explore the Layers →

02 - TCP-IP Protocol Stack

MOBILE COMPUTING 02 / 04

THE FOUR-LAYER MODEL

TCP/IP Protocol Stack

LOGICAL LAYERS Application → network link

Application Layer 04
HTTP/HTTPS, DNS, DHCP
Provides network services to applications.

Transport Layer 03
TCP, UDP
Supports end-to-end data transport.

Internet Layer 02
IP
Addresses and routes packets between networks.

Network Access Layer 01
Wi-Fi, Ethernet, cellular radio-link technologies
Transfers data across the local or wireless network link.

Each layer relies on services provided by lower layers.

← Previous

Next: Cellular Protocols →

03 - Cellular Protocol Stack

MOBILE COMPUTING 03 / 04

GSM / UMTS / LTE / 5G

Cellular Protocol Stack

Application and Internet Services

Web browsing • Messaging
Voice and video applications

Core Network Protocols

IP — Packet addressing and routing
TCP/UDP — Transport protocols used by applications
Signalling protocols — Manage connections and network procedures

TCP/UDP is end-to-end application transport; it does not terminate in the mobile core.

Radio Access and Link Protocols

Examples used in different cellular technologies:

GSM: Radio interface and GSM radio-link protocols

UMTS: Radio Resource Control (RRC), Packet Data Convergence Protocol (PDCP), Radio Link Control (RLC), and Medium Access Control (MAC)

LTE/5G: RRC, PDCP, RLC, MAC, and physical layer (PHY), with technology-specific differences



! Cellular protocol stacks differ by generation. The layers and protocols shown are representative examples, not one universal stack.

← Previous

View Data Flow →

04 - Mobile Data Communication Flow

MOBILE COMPUTING 04 / 04

A WEBSITE REQUEST, STEP BY STEP

Mobile Data Communication Flow

User Opens a Website

Application Layer
Creates an HTTP/HTTPS request.

Transport Layer
TCP or UDP carries the application data, depending on the protocol used.

Internet Layer
IP addresses and routes packets.

Cellular Radio Layers
RLC, MAC, and PHY carry data over the radio link, where applicable.

Base Station and Core Network
Forward the data towards the destination network.

Internet Server
Processes the request and returns data.

Response travels back to the device.

WORKING TOGETHER

Protocol layers work together to provide reliable, organised communication across mobile networks.

← Previous

Back to Start

01 - Protocol Stack Intr...
02 - TCP-IP Protocol St...
03 - Cellular Protocol St...
04 - Mobile Data Comm...

Result

A visual representation of mobile network protocol stacks was created in Figma, illustrating the five protocol layers, their functions, data-flow directions, and the relationship between Internet protocols and cellular network protocols.